

TB Step Shifter

DETAILED ENGLISH USER MANUAL

A harmonic-series pitch and color processor that moves selected overtones instead of simply resampling the entire spectrum.



Catalog version: Current

Documentation edition: 2026-08-04

Host-visible endpoints documented: 7

1. Product purpose

A harmonic-series pitch and color processor that moves selected overtones instead of simply resampling the entire spectrum.

Conventional pitch shifting moves every spectral component together. TB Step Shifter tracks a pitched source, identifies its harmonic series, and relocates selected harmonics. This allows pitch movement or upper-harmonic recoloring while preserving more of the original timing and, in Colorize mode, the fundamental.

What result it creates

- Harmonic-series transposition
- Upper-harmonic recoloring with a grounded fundamental
- Fast or fine analysis resolution
- Continuous dry/wet and harmonic selection control

Signal flow

Input -> pitch/fundamental analysis -> harmonic selection by Step -> Pitch relocation -> optional Colorize fundamental preservation -> reconstruction -> Mix

2. Complete feature guide

Pitch

Sets the destination shift from -24 to +24 semitones.

Step

Controls how much of the harmonic structure participates in the relocation. Lower values affect fewer components; higher values transform more of the series.

Colorize

When off, the detected harmonic series is shifted as a pitch effect. When on, the fundamental can remain grounded while upper harmonics move, creating a new timbral identity.

Resolution

Fast responds more quickly and suits changing notes; Fine uses more detailed analysis and suits stable, sustained material.

Mix and Programs

Mix blends the reconstruction with the original. Programs cover clean octave, fifth, deep octave, sub weight, harmonic glass, and warm patina.

3. Quick start

1. Use a clean, clearly pitched monophonic source.
2. Choose Fast for moving lines or Fine for sustained tones.
3. Set Pitch.
4. Raise Step until enough harmonics move.

5. Enable Colorize for timbral rather than full-note shifting.
6. Set Mix in context.

4. Practical workflows

Harmonic color

1. Enable Colorize.
2. Set Pitch to +7 or +12 semitones.
3. Use medium Step and Fine resolution.
4. Blend below full wet.

Expected result: The note center stays recognizable while overtones take on a glassy or shifted color.

Spectral octave

1. Disable Colorize.
2. Set Pitch to +12 or -12.
3. Increase Step and choose the resolution that follows the performance.

Expected result: The tracked harmonic series moves toward a distinct octave transformation.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Polyphonic or noisy sources can confuse fundamental tracking.
- Very low fundamentals may not provide enough stable harmonics.
- The effect is intentionally different from transparent broadband pitch shifting.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 7 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Colorize VST3 ID 1518668561	Off, On	Off	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Mix VST3 ID 108124	0.0% to 100%	100%	Automatable	Sets the balance between the original signal and the complete processed path.
Pitch VST3 ID 106677056	-24.0 st to +24.0 st	+12.0 st	Automatable	Moves pitch, frequency structure, or perceived resonant size for the named process.
Program VST3 ID 1886548852	Init, Clean Octave Up, Airy Fifth, Deep Octave, Sub Weight, Harmonic Glass, Warm Patina	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Resolution VST3 ID 547453100	Fast, Fine	Fast	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Step VST3 ID 3540684	0.0% to 100%	50%	Automatable	Sets how strongly the named process affects the signal.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.